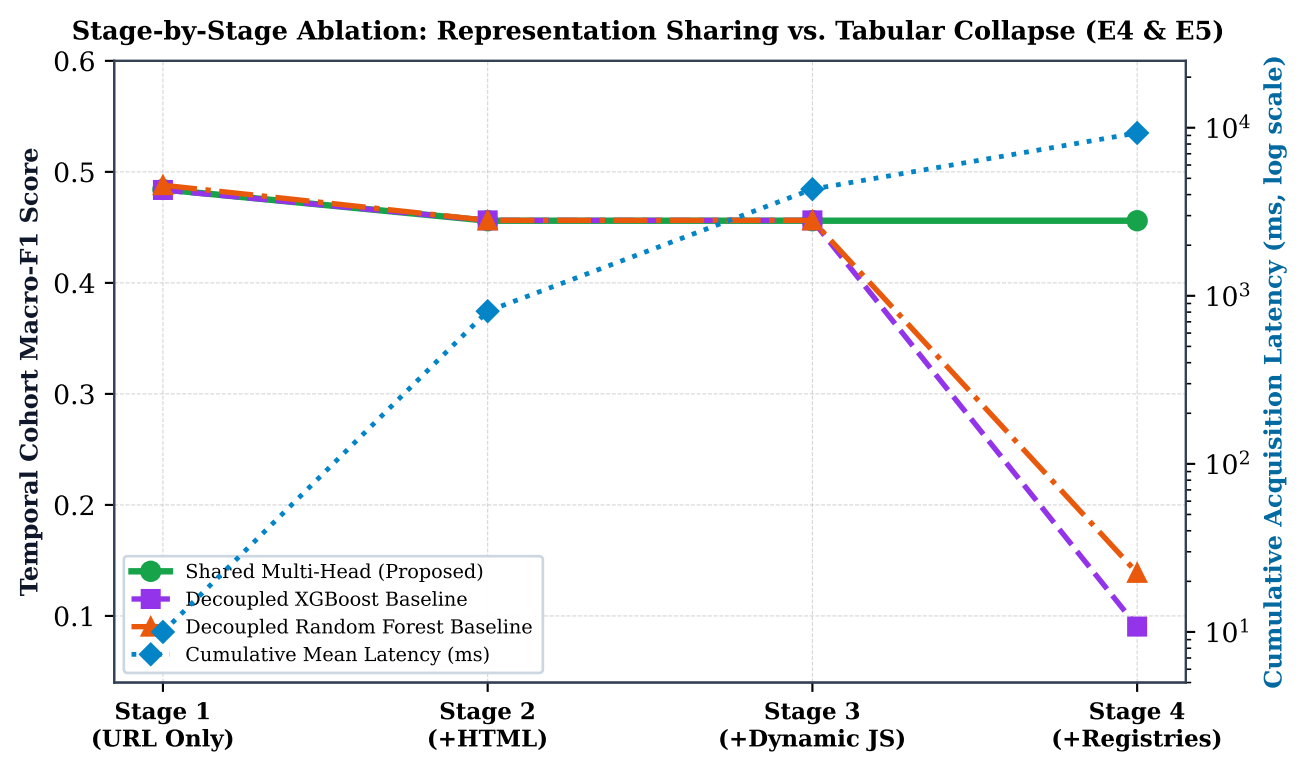
Stage-by-Stage Ablation: Representation Sharing vs. Tabular Collapse (E4 & E5)
Temporal Cohort Macro-F1 Score
Cumulative Acquisition Latency (ms, log scale)
0.6
0.5
0.4
0.3
0.2
0.1
10^4
10^3
10^2
10^1
Shared Multi-Head (Proposed)
Decoupled XGBoost Baseline
Decoupled Random Forest Baseline
Cumulative Mean Latency (ms)
Stage 1 (URL Only)
Stage 2 (+HTML)
Stage 3 (+Dynamic JS)
Stage 4 (+Registries)